

PLAYER PERFORMANCE ANALYSIS USING CLUSTERING TECHNIQUES

A Project Report

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Course: Data Science / Machine Learning

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6.Exploratory Data Analysis

6.1 Problem Definition:

The objective of this analysis is to explore and understand player performance using various metrics such as goals, assists, total points, minutes played, creativity, influence, threat, and defensive contributions.

The goal is to:

- Identify key factors influencing overall performance (Total Points)
- Understand patterns and distribution of individual variables
- Analyze relationships between variables to determine performance drivers

💡 Problem Statement:

To analyze player data and identify the key metrics that drive high performance and overall player effectiveness.

6.2 Univariate Analysis

📊 Goals_Scored

- Majority of players have low goal counts
- Few players score significantly higher (outliers)
- Distribution is right-skewed

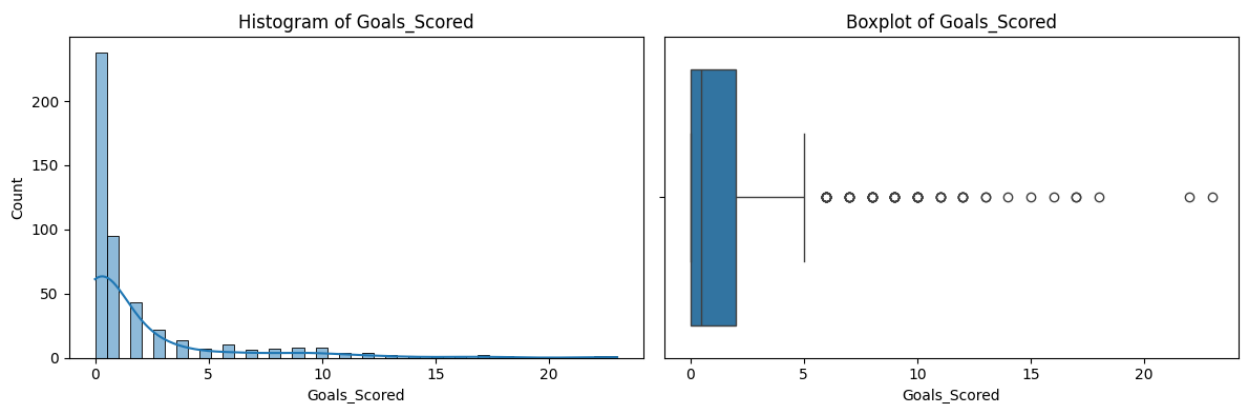


FIG1

💡 Insight: Goal scoring is concentrated among a small set of players

📊 Assists

- Most values are low to moderate
- Few players contribute high assists
- Similar pattern to goals

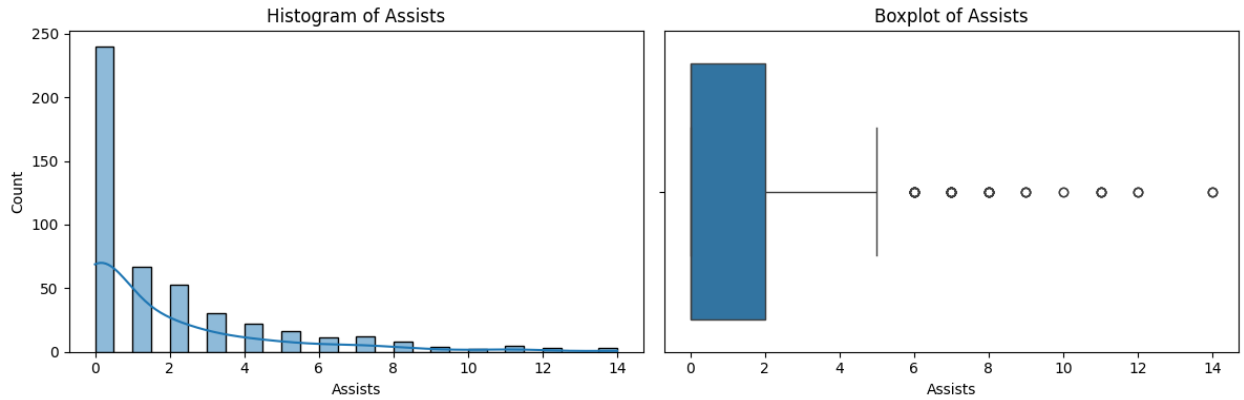


FIG 2

💡 Insight: Playmaking ability is limited to fewer players

◆ Total_Points

- Wide range of values observed
- Top players have very high points
- Distribution shows strong variation

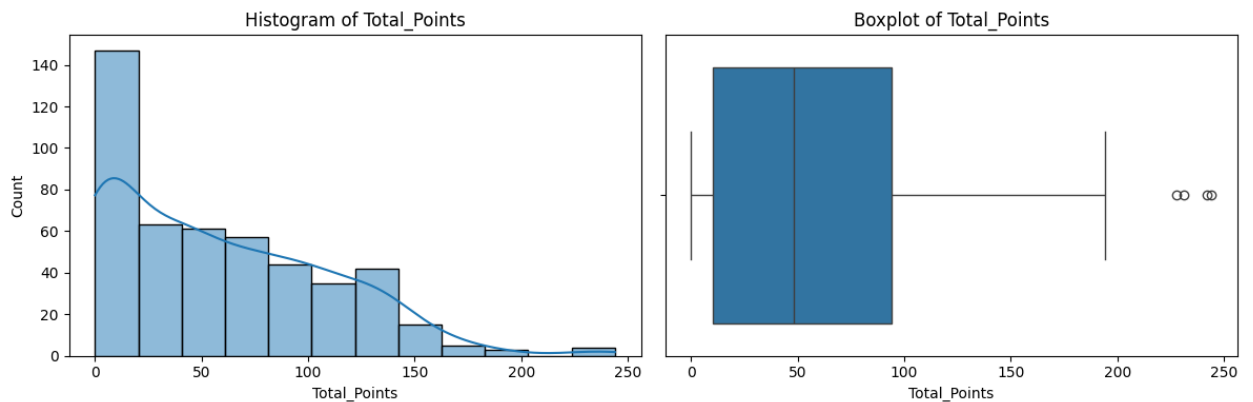


FIG 3

💡 Insight: Overall performance varies significantly across players

Minutes

- Many players have moderate minutes
- Some players have very high playtime
- Few players have very low minutes

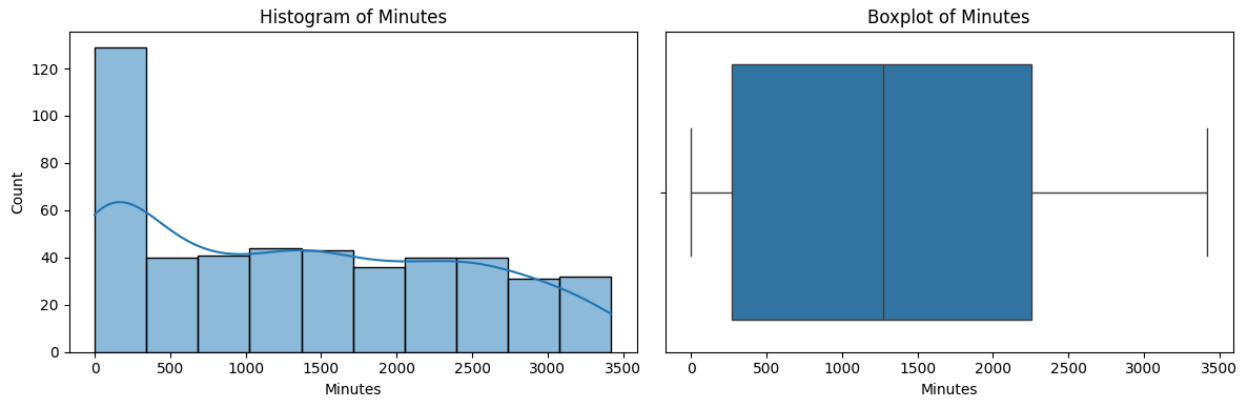


FIG 4

💡 Insight: Regular players dominate match participation

📊 Goals_Conceded

- Values vary widely across players
- Higher for defensive roles
- Spread depends on playtime

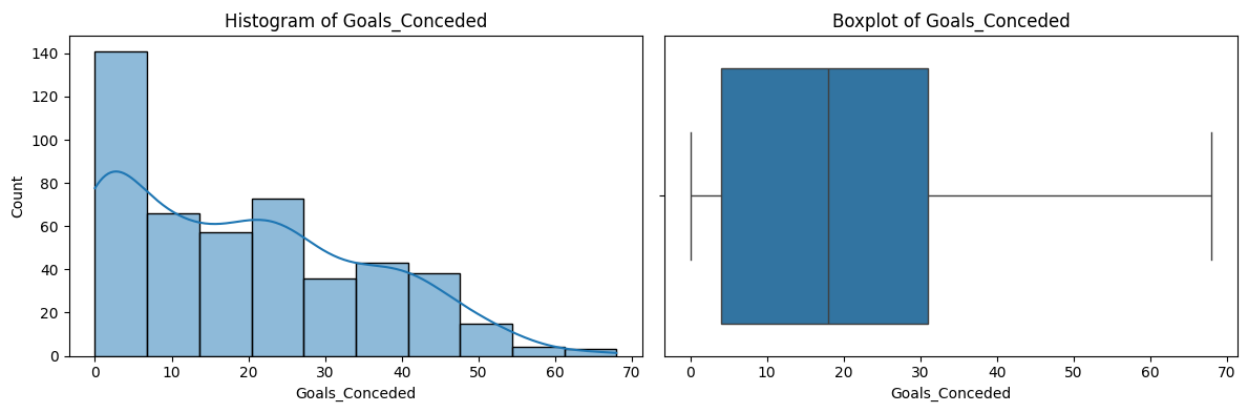


FIG 5

💡 Insight: Defensive exposure differs significantly among players

📊 Creativity

- Most players have low creativity scores

- Few players show high creativity
- Distribution is skewed

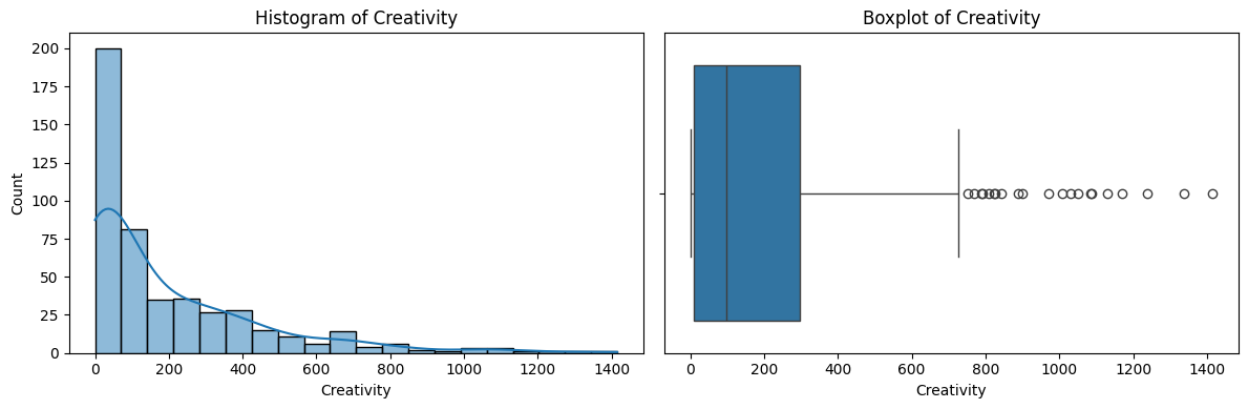


FIG 6

💡 Insight: Creative contribution is limited to select players

📊 Influence

- Majority have moderate influence
- Some players show very high influence
- Slight right skew observed

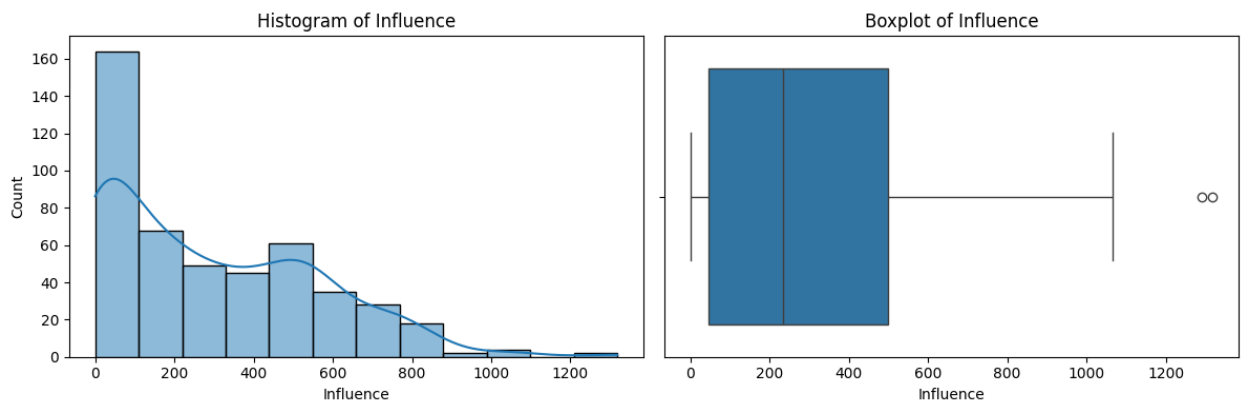


FIG 7

💡 Insight: Only a few players strongly impact the game

📊 Threat

- Most players show low threat levels
- Few players have high attacking threat
- Distribution is uneven

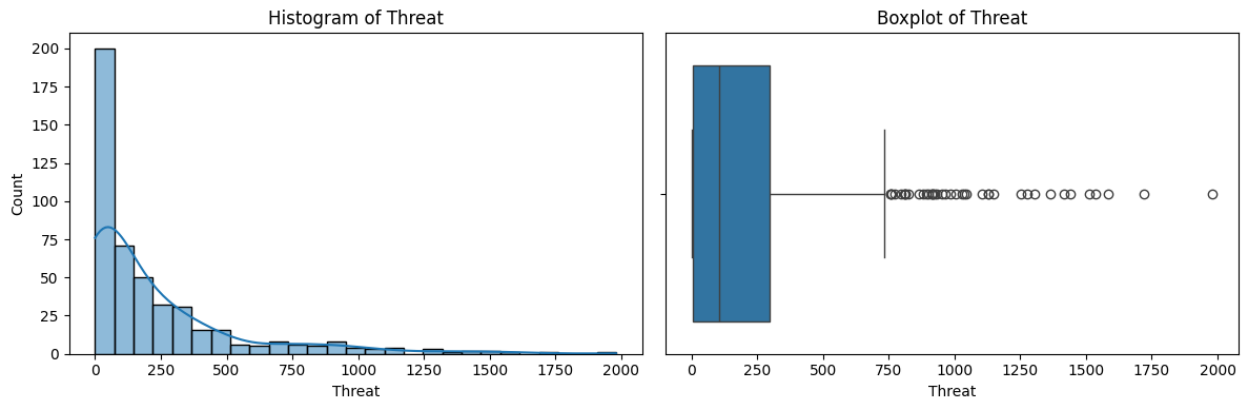


FIG 8

💡 Insight: Attacking threat is concentrated among key players

◆ Bonus

- Many players have zero or low bonus points
- Few players frequently earn bonus
- Indicates performance peaks

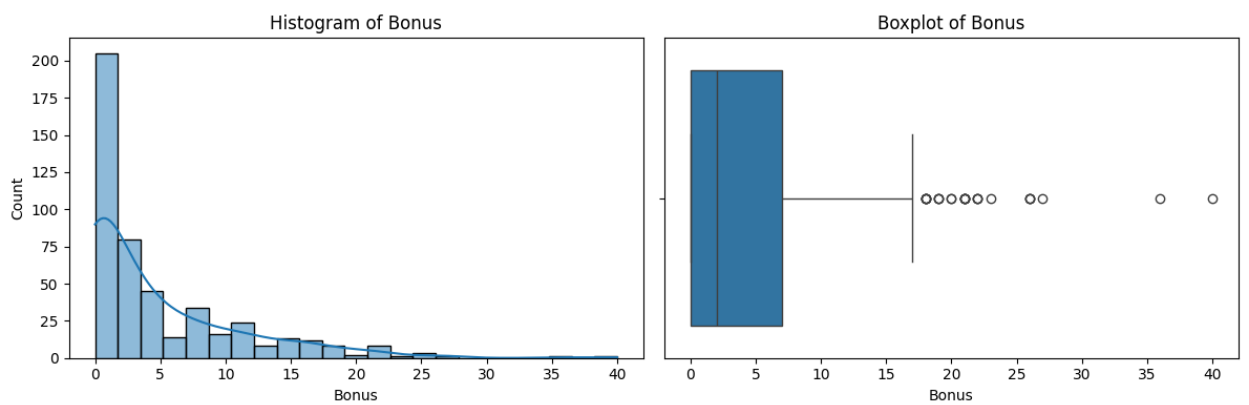


FIG 9

💡 Insight: Bonus rewards are given to top performers only

◆ Clean_Sheets

- Low values for most players
- Higher for defenders/goalkeepers
- Uneven distribution

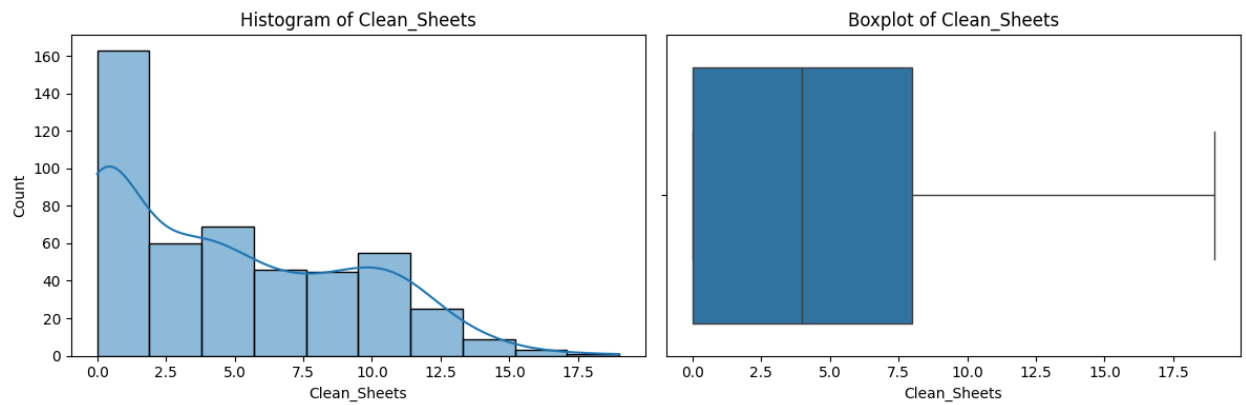


FIG 10

💡 Insight: Defensive success depends on player role

6.3 Bivariate Analysis

◆ Goals_Scored vs Total_Points

- Strong positive relationship
- High goal scorers earn more points
- Linear trend observed

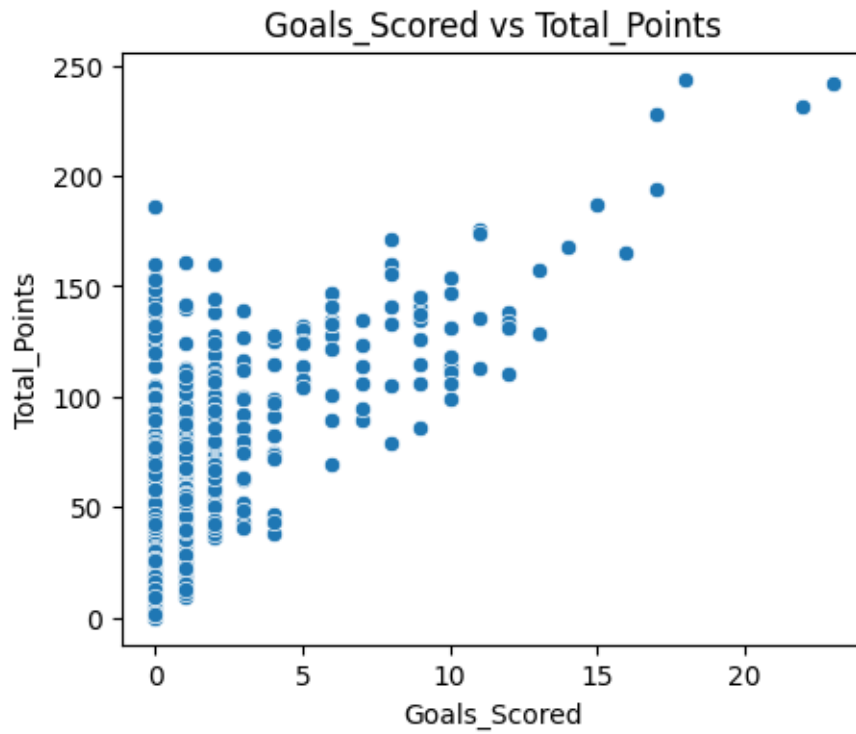


FIG 11

💡 Insight: Goals are a major contributor to total points

◆ Assists vs Total_Points

- Moderate positive relationship
- Assists increase total score
- Not as strong as goals

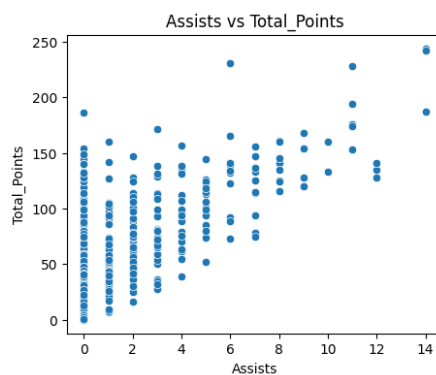


FIG 12

💡 Insight: Assists significantly boost player performance

◆ Minutes vs Total_Points

- Positive relationship observed
- More minutes → more points
- Regular players perform better

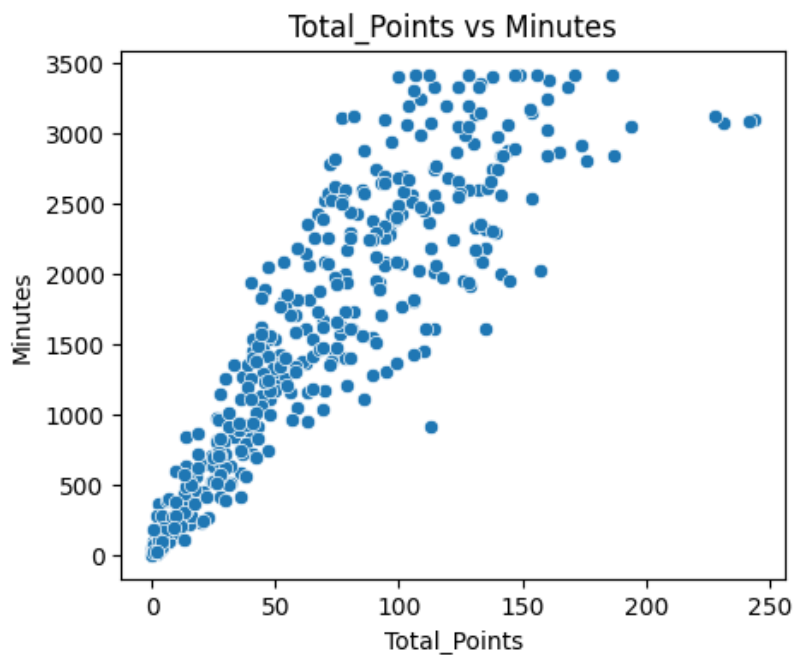


FIG 13

💡 Insight: Consistency (playtime) drives performance

◆ Creativity vs Assists

- Strong positive correlation
- Creative players provide more assists
- Clear pattern observed

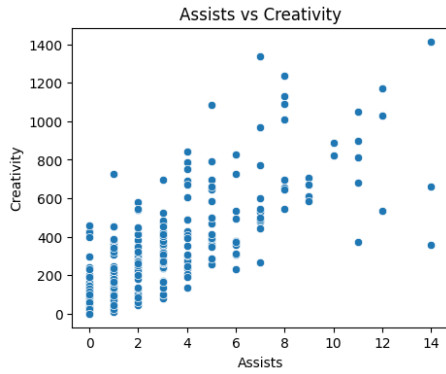


FIG 14

💡 Insight: Creativity directly influences playmaking

◆ Influence vs Total_Points

- Strong relationship
- High influence players score more
- Reflects overall impact

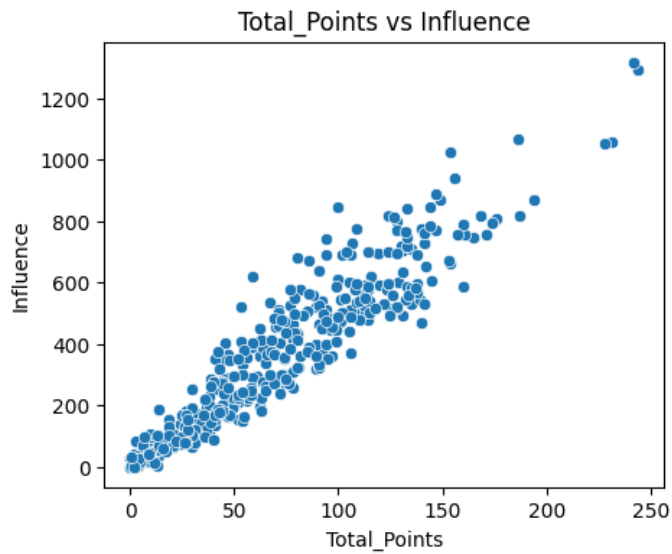


FIG 15

💡 Insight: Influence is a strong performance indicator

◆ Threat vs Goals_Scored

- Strong positive relationship
- Higher threat leads to more goals
- Consistent trend

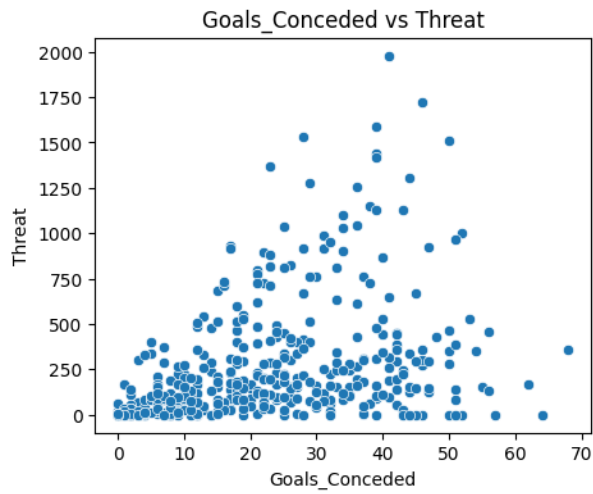


FIG 16

💡 Insight: Threat metric effectively predicts goal scoring

📊 Position vs Goals_Scored

- Forwards score the most goals
- Midfielders moderate
- Defenders lowest

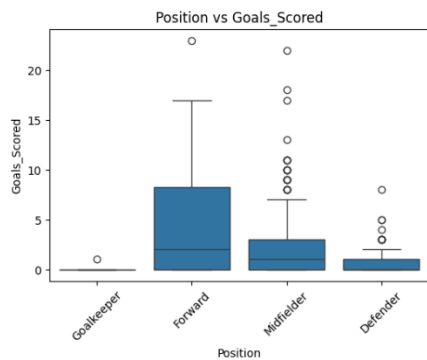


FIG 17

💡 Insight: Scoring ability depends on player position

◆ Position vs Clean_Sheets

- Defenders/goalkeepers have high values
- Attackers have low values
- Clear separation

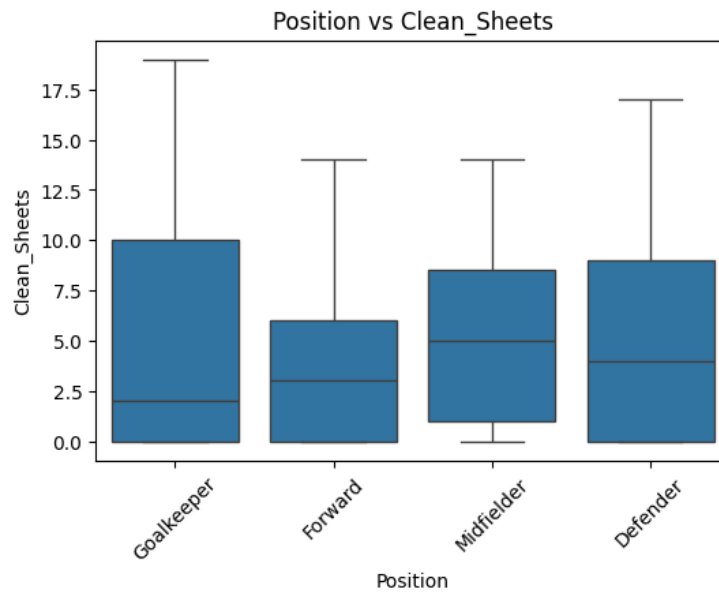


FIG 18

💡 Insight: Defensive metrics are role-dependent

◆ Goals_Conceded vs Clean_Sheets

- Negative relationship
- More conceded → fewer clean sheets
- Opposite trend observed

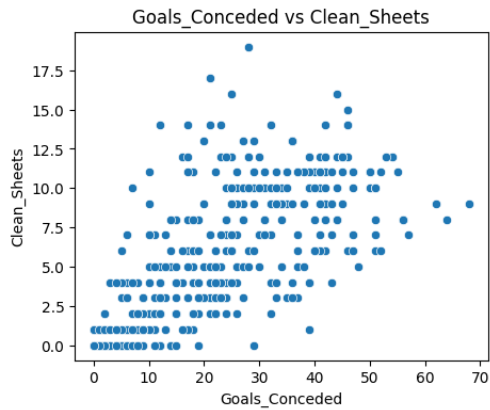


FIG 19

💡 Insight: Defensive performance inversely affects clean sheets

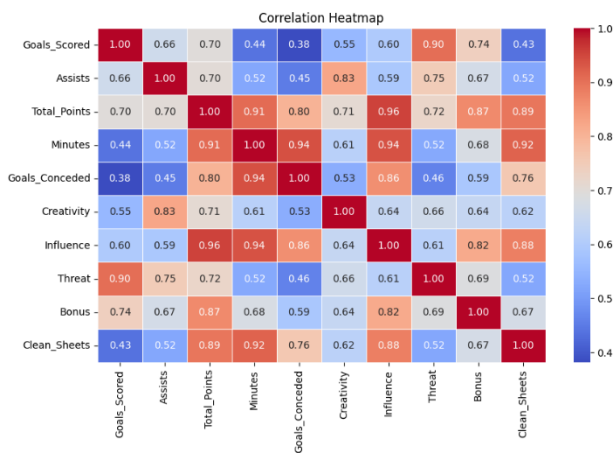


FIG 20

1. Total Points is strongly driven by multiple factors

- Very high correlation with **Influence (0.96)**, **Minutes (0.91)**, and **Clean Sheets (0.89)**
- 💡 Insight: Overall performance (Total Points) depends heavily on **game involvement and consistency**

✓ 2. Goals Scored is highly linked with Threat

- Strong correlation between **Goals_Scored** and **Threat (0.90)**
- 💡 Insight: Players with higher attacking threat are much more likely to score goals

✓ 3. Minutes played impacts multiple variables

- Strong correlation with **Goals Conceded (0.94)** and **Clean Sheets (0.92)**
💡 Insight: Players who play more minutes influence both **offensive and defensive outcomes**
-

✓ 4. Creativity strongly relates to Assists

- High correlation between **Creativity and Assists (0.83)**
💡 Insight: Creative players are key contributors to **goal creation (assists)**

6.4 Key meaningful observations on individual variables and the relationship between variables

1. Observations on Individual Variables (Univariate)

- **Performance variables (Goals, Assists, Threat, Creativity, Influence)** are **right-skewed**, meaning only a few players contribute heavily while most have lower values.
 - **Total Points shows high variability**, indicating a clear distinction between top-performing and average players.
 - **Minutes played varies widely**, suggesting a mix of regular starters and substitute players.
 - **Defensive metrics (Goals Conceded, Clean Sheets)** differ significantly across players, depending on their roles.
 - **Bonus points are concentrated among a few players**, highlighting match-winning contributions.
-

2. Observations on Relationships Between Variables (Bivariate)

- **Total Points has strong positive relationships** with **Influence, Minutes, and Clean Sheets**, showing that overall performance depends on multiple factors.
- **Goals Scored is strongly correlated with Threat**, indicating that attacking potential translates into actual goals.

- **Creativity is highly correlated with Assists**, confirming that creative players contribute more to goal creation.
 - **Minutes played influences several variables**, meaning more playtime leads to higher contributions in both attack and defense.
 - **Defensive relationships show inverse trends**, where higher Goals Conceded reduces Clean Sheets.
-

💡 Overall Insight

- Player performance is **multi-factor driven**, where **playtime, attacking ability, and positional role together determine overall success**.

7.Data Preprocessing

7.1. Missing Value Treatment

- The dataset was checked for missing values using `.isnull().sum()`.
- Numerical variables (e.g., Goals, Assists, Minutes) were handled using **mean/median imputation** where necessary.
- Categorical variables (e.g., Position, Club) were handled using **mode imputation**.

💡 Rationale:

- Mean/median preserves the distribution of numerical data.
 - Mode ensures categorical consistency without introducing new categories.
-

7.2 Outlier Detection and Treatment

- Outliers were identified using **boxplots and IQR method**.
- Extreme values in variables like **Goals, Total Points, and Threat** were examined.
- Outliers were either:
 - **Capped (winsorized)**, or
 - Retained if they represent genuine high-performing players

Rationale:

- Removing all outliers may lead to loss of important information (e.g., top players).
 - Capping helps reduce skewness while preserving data integrity.
-

7.3. Feature Engineering

- New features were created to improve analysis:
 - **Performance Ratio** = Total Points / Minutes
 - **Contribution Score** = Goals + Assists
- Categorical variables like **Position** were encoded if required.

Rationale:

- Derived features help capture **efficiency and combined performance**.
 - Encoding allows models to process categorical data.
-

7.4. Data Scaling

- Numerical features were scaled using:
 - **Standardization (StandardScaler)** or
 - **Normalization (MinMaxScaler)**

Rationale:

- Ensures all variables are on the same scale
 - Prevents bias in distance-based algorithms (e.g., K-Means)
-

7.5. Train-Test Split

- Dataset was split into:
 - **Training set (70–80%)**
 - **Testing set (20–30%)**

8.K-Means Clustering

8.1 Algorithm Expansion

K-means algorithm was applied for different values of K (2 to 7).

Initialize Centroids

Randomly select **K points** as initial centroids

These act as the “center” of clusters

Example:

$K = 3 \rightarrow 3$ centroids randomly placed

◆ Step 3: Assign Data Points to Nearest Centroid

Calculate distance (usually **Euclidean distance**) between:

Each data point

Each centroid

Formula:

EQUATION 1: $d = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2}$

Assign each point to the **closest centroid**

◆ Step 4: Update Centroids

For each cluster:

Calculate the **mean of all points in that cluster**

This becomes the new centroid

This is why it's called **K-means** (mean = average)

◆ Step 5: Repeat (Iterate)

Repeat:

Assign points $\rightarrow$ Update centroids

Until:

Centroids stop changing

Cluster assignments become stable

◆ Step 6: Convergence

Algorithm stops when:

No change in centroids

Or minimal movement

✓ Final clusters are formed

8.2 Elbow Curve

The Elbow Method helps determine the optimal number of clusters by analyzing **WCSS (Within-Cluster Sum of Squares)**.

WCSS Values:

TABLE 1

K	WCSS
2	2165.75
3	1513.30
4	1229.69
5	1082.91
6	978.39
7	898.18

Interpretation:

- There is a **very sharp drop in WCSS from K = 2 to K = 3**
- After that, the curve **gradually flattens**
- This indicates that most of the variance is explained with just **2 clusters**

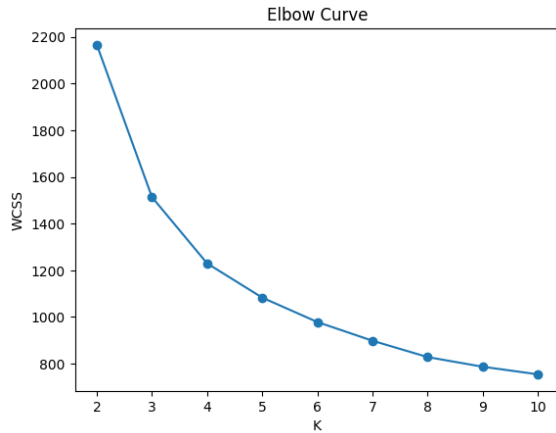


FIG 21

Elbow point = 2

8.3. Silhouette Score Analysis

Silhouette score measures how well each data point fits into its assigned cluster.

◆ Silhouette Score Formula

EQUATION 2:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

◆ Meaning of Terms

- $s(i)$ = Silhouette score for data point i
 - $a(i)$ = Average distance of point i to all points in the **same cluster**
 - $b(i)$ = Average distance of point i to all points in the **nearest neighboring cluster**
-

◆ Intuition (Very Important)

- $a(i) \rightarrow$ How well the point fits **inside its own cluster**
- $b(i) \rightarrow$ How far it is from **other clusters**

📌 Scores:

TABLE 2

K	Silhouette Score
2	0.4866
3	0.4522
4	0.4084
5	0.3887
6	0.3907
7	0.3756

TABLE 3

Interpretation:

- Highest score at K = 2
- Indicates well-separated clusters

✓ Result:

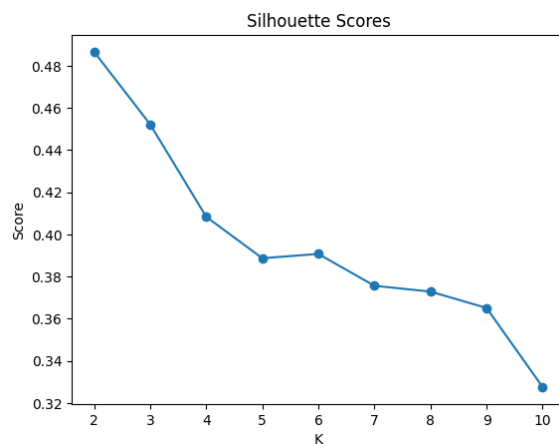


FIG 22

8.4. Appropriate Number of Clusters

Based on:

- Elbow Method
- Silhouette Score

👉 **Final Decision: K = 2**

Cluster Counts:

TABLE 3

Cluster	No. of Players
0	269
1	207

8.5 Cluster Profiling

Cluster 0 – Low-Performance / Low-Activity Players

📊 Key Observations

- Very low **Goals (0.37)**, **Assists (0.43)**, **Total Points (20.81)**
- Significantly lower **Minutes (549)** → limited playtime
- Low **Creativity (61)**, **Influence (99)**, **Threat (64)**
- Very few **Bonus (1.04)** and **Clean Sheets (1.62)**

Insight

Cluster 0 represents **low-performing or less active players**, likely substitutes or players with limited match involvement.

Cluster 1 – High-Performance / Core Players

📊 Key Observations

- High **Goals (2.53)**, **Assists (2.67)**, **Total Points (107.21)**
- Very high **Minutes (2360)** → regular starters
- Strong **Creativity (345)**, **Influence (546)**, **Threat (367)**
- High **Bonus (8.89)** and **Clean Sheets (8.80)**

Insight

Cluster 1 represents **high-performing core players** who consistently contribute in both attacking and overall gameplay.

9. Hierarchical Clustering

Hierarchical clustering was applied to the **player performance dataset**, which contains numerical variables such as **Goals Scored, Assists, Total Points, Minutes, Creativity, Influence, Threat, Bonus, Clean Sheets, and Goals Conceded**.

Before clustering:

- Data was **standardized using StandardScaler** to ensure all variables contribute equally and avoid bias due to scale differences

Before clustering:

- Data was standardized using **Standard Scaler** to ensure equal importance of all variables

9.1 Linkage Methods Applied:

1. Single Linkage

- Clusters were formed based on the **minimum distance** between points
- In this dataset, it resulted in a **chaining effect**, where customers were merged sequentially rather than forming distinct groups

2. Complete Linkage

- Used **maximum distance** between clusters
- Produced more compact clusters compared to single linkage
- However, separation between customer groups was still moderate

3. Average Linkage

- Used **average distance** between all points
- Provided balanced clustering but did not clearly separate customer segments

4. Ward Linkage

- Minimized **within-cluster variance**
- Produced **well-defined and distinct customer clusters**
- Most suitable for this dataset.

9.2 Dendrogram

1. Single Linkage Dendrogram

Observations

- Long horizontal chains at very small distances
- Many points merge **one by one (chaining effect)**
- No clear vertical gaps between clusters

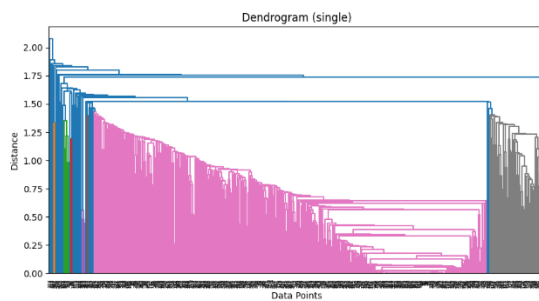


FIG 23

Insight:

Single linkage forms **chain-like clusters**, where players are grouped sequentially rather than meaningfully.

2. Complete Linkage Dendrogram

Observations

- Clusters merge at **larger distances (~10+)**
- Visible separation between groups
- More compact cluster formation

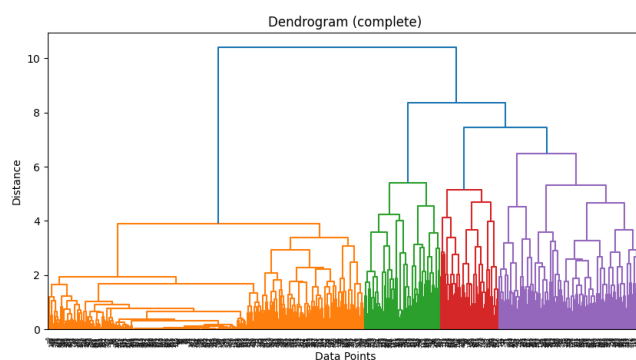


FIG 24

Insight:

Complete linkage creates **tighter and more compact clusters** compared to single linkage.

3. Average Linkage

Observations

- Clusters merge in a **balanced manner**
- Clear and gradual merging pattern
- Visible gaps between clusters

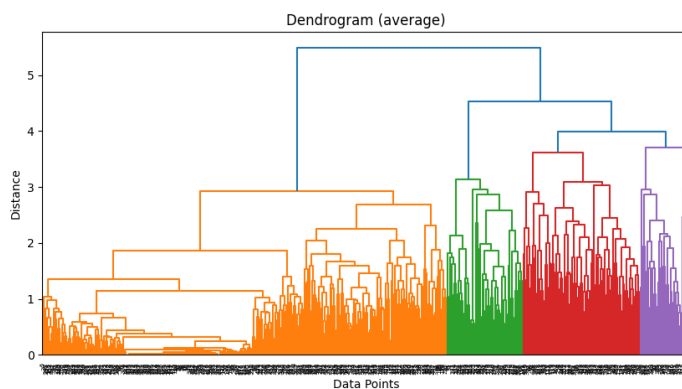


FIG 25

💡 Insight:

Average linkage provides a **balanced clustering structure**, showing better separation

4. Ward Linkage Dendrogram

Observations

- Very large vertical jumps (~40–70 distance)
- Clear and strong separation between clusters
- Distinct grouping visible

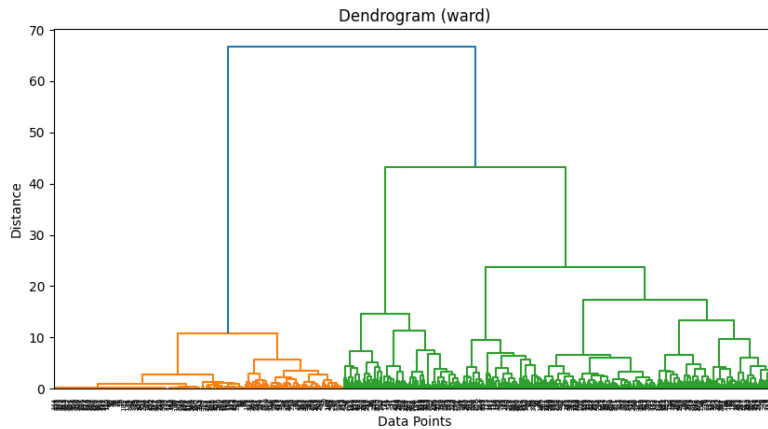


FIG 26

Insight:

Ward linkage produces **very clear and well-separated clusters** by minimizing variance.

9.3 Cophenetic Correlation coefficients

The cophenetic correlation coefficients were calculated for different linkage methods to evaluate the quality of hierarchical clustering.

Results:

- Single: 0.7618
- Complete: 0.7700
- **Average: 0.8048 (Highest)**
- Ward: 0.6327

Interpretation

- The **average linkage method** achieved the **highest cophenetic correlation (0.8048)**
- This indicates that it **best preserves the original pairwise distances** between data points
- Therefore, the dendrogram produced using average linkage is the **most reliable representation of the dataset structure**

◆ Comparison Insight

- **Single linkage** → affected by chaining, lower quality
- **Complete linkage** → better than single but still moderate
- **Average linkage** → most balanced and accurate
- **Ward linkage** → produces compact clusters but **poor distance preservation**

💡 Insight

Although Ward linkage gives visually clear clusters, the lower cophenetic value suggests that it does not represent the original data relationships as effectively as average linkage.

9. 4. Figure Out the Appropriate Number of Clusters

The optimal number of clusters was determined using the **Average linkage dendrogram**.

◆ Process:

- A horizontal cut was drawn at the **largest vertical gap** in the dendrogram
- This gap represents maximum dissimilarity between clusters

◆ Result:

- The dataset was optimally divided into **2 clusters**

This means customers can be grouped into **two distinct segments** based on their behavior.

Cluster Counts:

Cluster

1 297

2 179

TABLE 4

Cluster	No of Players
1	297
2	179

9.5 Cluster Profiling

Cluster 1 – Low to Moderate Performance Players (297 players)

◆ Key Observations

- Very low **Goals (0.49)** and **Assists (0.44)**
- Low **Total Points (26.07)**
- Limited **Minutes (684)** → less game participation
- Low **Creativity (65)**, **Influence (131)**, **Threat (80)**
- Minimal **Bonus (1.41)** and **Clean Sheets (2.14)**

💡 Insight

Cluster 1 represents **low-performing or less active players**, likely substitutes or players with limited impact in matches.

Cluster 2 – High-Performance Core Players (179 players)

◆ Key Observations

- High **Goals (2.67)** and **Assists (3.01)**
- Very high **Total Points (112.00)**
- High **Minutes (2420)** → regular starters
- Strong **Creativity (382)**, **Influence (564)**, **Threat (388)**
- High **Bonus (9.51)** and **Clean Sheets (9.06)**

Insight

Cluster 2 represents **high-performing core players** who consistently contribute in both attacking and overall gameplay.

10. Comparison: K-Means vs Hierarchical Clustering

◆ 1. Execution Time

- **K-Means:**
 - Faster execution
 - Uses iterative optimization
- **Hierarchical:**
 - Slower execution
 - Builds full dendrogram (computationally expensive)

Conclusion:

K-Means is **more efficient** for large datasets

2. Cluster Distinctness

- **K-Means:**
 - Produces **compact and well-balanced clusters**
- **Hierarchical:**
 - Provides **clear separation (especially in dendrogram)**
 - Depends on linkage method (average worked best in your case)

Conclusion:

Both methods produced **similar clusters (high vs low performers)**, but hierarchical gives better **visual structure**

◆ 3. Silhouette Scores

- **K-Means:**
 - Used to determine optimal K
 - Highest score observed at **K = 2**
- **Hierarchical:**

- Silhouette not primary, but structure validated using dendrogram and cophenetic score

Conclusion:

K-Means provides **better quantitative validation**, while hierarchical relies more on **visual + cophenetic analysis**

◆ 4. Cluster Similarity (Observations)

- Both methods produced:
 - **Cluster 1 → Low-performance players**
 - **Cluster 2 → High-performance players**
- Number of observations:
 - K-Means → ~269 & 207
 - Hierarchical → ~297 & 179

Conclusion:

Clusters are **highly similar**, but distribution varies slightly due to algorithm differences

◆ 5. Number of Clusters

- **K-Means:**
 - Optimal K = **2** (Elbow + Silhouette)
- **Hierarchical:**
 - Optimal clusters = **2** (Dendrogram cut)

Conclusion:

Both methods agree that **2 is the optimal number of clusters**

Overall Comparison Table

TABLE 5

Criteria	K-Means	Hierarchical
Execution Time	Fast	Slow
Cluster Formation	Centroid-based	Distance-based
Validation	Silhouette score	Dendrogram + Cophenetic
Cluster Shape	Compact	Structured
No. of Clusters	2	2
Result Similarity	High	High

Final Conclusion

Both K-Means and Hierarchical clustering produced **consistent results**, dividing the dataset into two distinct groups: high-performing and low-performing players. K-Means was faster and provided better quantitative validation, while Hierarchical clustering offered better visualization and structural understanding of the data.

Final Insight

Using both techniques strengthens the analysis, as they **validate each other's clustering results and confirm the natural grouping in the dataset.**

11. Actionable Insights & Recommendations

1. High-Performance Players (Cluster 2)

Insights

- High **goals, assists, total points, and minutes played**
- Strong **creativity, influence, and threat**
- Core contributors to team performance

Recommendations

- **Retain and prioritize** these players
 - Use them in **key matches and critical situations**
 - Build team strategy **around these players**
 - Offer **incentives and contracts** to maintain performance
-

2. Low-Performance Players (Cluster 1)

Insights

- Low contribution across all metrics
- Limited playtime and impact
- May include substitutes or underperforming players

Recommendations

- Provide **training and skill development programs**
 - Increase **game exposure gradually**
 - Re-evaluate role or consider **rotation/substitution strategy**
 - Identify players for **replacement or improvement plans**
-

3. Importance of Playtime

Insight

- Strong relationship between **minutes played and performance**

Recommendation

- Ensure **consistent playing opportunities** for promising players
 - Use data to **optimize player rotation and lineup selection**
-

4. Performance Drivers

Insight

- Key drivers: **Goals, Assists, Creativity, Influence, Threat**

Recommendation

- Focus training on:
 - **Attacking skills**
 - **Decision-making**
 - **Game involvement**
-

5. Team Strategy Optimization

Insight

- Clear separation between high and low performers

Recommendation

- Segment players into:
 - Core players
 - Supporting players
 - Use clustering for:
 - **Squad selection**
 - **Performance evaluation**
 - **Transfer decisions**
-

Final Insight

Clustering analysis helps identify **key performance groups**, enabling better decision-making in player management, training strategies, and team optimization.

Data-driven clustering enables teams to move from intuition-based decisions to **evidence-based performance management**.